Design, Modeling, and Experimental Assessment of Heat-Exchanger-Based Reflux Batch Distillation System for Ethanol-Water Separation

Saroj Basnet[1], Sachin Saud[1], Sumi Shakya[2], Laxman Motra[1], Bikki Chhantyal[3]*
[1]Department of Mechanical and Aerospace Engineering, Pulchowk Campus, Institute of Engineering, Tribhuvan University, Nepal

[2]Department of Applied Sciences and Chemical Engineering, Pulchowk Campus, Institute of Engineering, Tribhuvan University, Nepal

[3]Department of Automobile and Mechanical Engineering, Thapathali Campus, Institute of Engineering, Tribhuvan University, Nepal

Email: chhantyalbikki@tcioe.edu.np

ABSTRACT

While conventional batch distillation relies on direct flow measurements for reflux control, laboratory and microscale units generally lack these sensors, necessitating alternative methods to estimate process dynamics. This study presents the design, modeling, fabrication, and experimental testing of a heat-exchanger-based reflux batch distillation system for ethanol–water separation. The system uses a dual-condenser arrangement, and effective heating power and reflux ratio are estimated from heat-up and condenser energy balances without direct flow measurements. The model incorporates the local atmospheric pressure in Kathmandu, Antoine vapor-pressure correlations, the Non-Random Two-Liquid (NRTL) activity-coefficient model, McCabe–Thiele stage analysis, and a time-discretized equilibrium-stage solver implemented in Python. The fabricated three-tray bubble-cap column was tested in three batch runs using initial ethanol concentrations of 20%, 27%, and 30% (v/v). For the primary experimental case (Run 3), the model predicted a final distillate concentration of 83.70% (v/v) and a collected volume of 1094.49 mL, compared with experimental values of 79% and 1049 mL, corresponding to deviations of 5.62% and 4.16%, respectively. Across the three runs, the concentration and volume normalized root-mean-square error (NRMSE) values were approximately 6.66% and 7.83%, respectively. The model predictions also showed close agreement with Aspen Plus results, with deviations below 7.3% for the quantities compared. A controlled pressure comparison showed that changing from sea-level to Kathmandu pressure at fixed heating power and reflux ratio altered the predicted distillate composition and total amount by less than 0.5%. The results demonstrate that the developed energy-balance approach and open numerical model can provide useful predictions of batch distillation behavior using relatively simple instrumentation, while also providing a practical basis for further development of small-scale distillation systems.

Keywords: *Batch distillation; Ethanol–water separation; Reflux distillation; NRTL; Dynamic modeling; McCabe–Thiele; Small-scale distillation.*

1. INTRODUCTION

Distillation is a fundamental separation process widely used in bioethanol production and pharmaceutical applications to obtain high-purity solvents and intermediates required for fuel blending and drug production [1]. A key challenge in ethanol–water separation by conventional distillation is the presence of a minimum-boiling azeotrope at ~95.6 wt% ethanol at 1 atm, which prevents the production of absolute ethanol. Even with this limitation, conventional distillation is typically the favored technology in most process industries due to its reliable and well-understood nature [1, 2]. The strategic importance of improving separation efficiency has been widely recognized in the chemical engineering literature [2], and Sorensen [3] has comprehensively treated the principles of batch distillation design and operation. Continuous distillation units are typically employed in large-scale bioethanol production facilities as they offer high throughput, consistent output, and stable performance. Conversely, batch distillation units, with higher upfront costs, are better suited for pilot plants, research and development laboratories, pharmaceutical laboratories, and small-scale bioethanol and solvent recovery plants where flexibility in feed compositions is a critical consideration [4].

Although batch distillation is a widely used technology in small-scale facilities, many existing batch distillation units in laboratories are still built on engineering intuition or experience [5]. In most cases, such systems lack detailed thermodynamic considerations or robust experimental verification. As a result, these systems suffer from suboptimal energy utilization, inconsistent product quality, and insufficient predictive capacity for process improvement [6].

Several modeling studies have highlighted the need to combine rigorous modeling with experimental verification in batch distillation systems. For example, Bonsfills and Puigjaner [7] formulated a dynamic model of a batch distillation column that was rigorously validated through experiments, whereas Alhaboubi [8] found that employing variable reflux operation allows for maintaining near-constant distillate composition with higher energy efficiency than fixed reflux operation. Nonetheless, most published works assume operation at sea-level atmospheric pressure and in industrial or pilot-scale settings. Operating at high-altitude (above 1000 m) lowers atmospheric pressure. Reduced pressure significantly alters the boiling points of ethanol and water and drastically changes their phase-equilibrium behavior [9]. Guevara Luna et al. [10] showed that parameter values obtained from Vapor–Liquid Equilibrium (VLE) for the binary system ethanol–water at standard pressure are not directly applicable under low-pressure conditions. Despite its impact, the influence of operating pressure is rarely addressed in laboratory-scale batch distillation models.

1.1 Modeling and Optimization of Batch Distillation

Classical distillation relies on equilibrium-stage methods such as the Rayleigh equation and the McCabe–Thiele diagram, which serve as basic tools for evaluating the required separation and the number of stages [11]. These methods are still commonly used in small-scale design and preliminary analysis. However, such approaches are limited in their ability to capture the dynamic nature of batch distillation processes [12]. More rigorous stage-wise dynamic models offer improved prediction accuracy, but their computational complexity makes them less practical for laboratory and decentralized systems. As a result, reduced-order and equilibrium-based dynamic models have been widely adopted. Mujtaba and Macchietto [4]

showed that equilibrium-stage formulations can be transformed into time-discretized models that track the evolution of composition while maintaining mass-balance consistency, and Luyben [13] demonstrated the effectiveness of simplified dynamic models.

Accurate thermodynamic data for VLE are crucial for describing batch distillation performance [14]. In particular, for ethanol–water systems, Non-Random Two-Liquid (NRTL) and Universal Quasi-Chemical (UNIQUAC) models are widely employed because they account for non-ideal liquid behavior using activity-coefficient models based on local composition and excess Gibbs energy concepts [15–17]. For binary systems at moderate pressures, a simpler model such as the Margules equation can provide adequate accuracy with low computational overhead, making it suitable for use in reduced-order dynamic models [18, 19].

Few studies have examined the influence of operating pressure within modeling frameworks. Do et al. [20] highlighted the importance of accounting for pressure variations in distillation performance and energy requirements, while Schwarz [15] emphasized that inconsistent VLE datasets can lead to inaccurate predictions. The mode of operation also significantly influences batch distillation performance [21]. In fixed-reflux operation, the distillate composition varies continuously over time, whereas variable-reflux strategies improve product consistency and energy efficiency. Alhaboubi [8] demonstrated that variable-reflux operation enables near-constant distillate composition while improving energy utilization.

The use of surrogate and machine learning-based models for distillation columns has gained attention recently. Bubel et al. [22] developed a set of reusable surrogate models for distillation columns in Aspen HYSYS. Chen et al. [23] developed physics-informed neural networks that learn to model the complex thermodynamics of multicomponent chemical systems from real process data for batch distillation. However, these techniques require large amounts of data for training and model tuning, which are typically unavailable in small laboratory settings.

For custom-built, laboratory-scale columns, performance is further affected by heat losses and instrumentation deficiencies [24]. Ränger et al. [25] conducted an experimental investigation of thermal losses for pilot-scale batch distillation units, and Duan et al. [26] quantified the effect of operating pressure on the required heating and cooling duties of an industrial-scale system. These factors show the importance of addressing both energy balance and operating pressure in models of small-scale distillation columns. Despite considerable advances, most published studies assume standard atmospheric pressure and use commercial simulators. This approach offers limited insight into optimizing small-scale distillation units in locations under nonstandard pressure and temperature conditions.

1.2 Pressure Sensitivity and High-Altitude Effects

Standard procedures for distillation design, regardless of the application, have commonly assumed sea-level atmospheric pressure (101.325 kPa) [13]. The atmospheric pressure at high altitudes (e.g., Kathmandu, Nepal, approximately 1300 m above sea level) is lower than at sea level, which in turn directly affects the boiling temperature of liquids and the VLE. Battisti et al. [27] showed that pressure variation can enhance separation in pressure-swing distillation. Guevara Luna et al. [10] demonstrated that VLE data obtained under atmospheric conditions are not applicable at low pressure and would result in significant errors. Moreover, Duan et al. [26]

reported the effect of operating pressure on the energy duties needed to run the distillation column.

Although these studies underline the importance of pressure effects, they primarily focus on industrial systems, with limited attention given to laboratory-scale batch distillation under naturally reduced-pressure conditions.

1.3 Experimental Validation in Small-Scale Units

Although substantial progress has been made in modeling and simulation, actual experimental studies using low-cost, locally available materials and equipment remain scarce. Rougier et al. [28] investigated thermal losses and their impact on small-scale batch distillation, concluding that they play a significant role. Barolo and Botteon [29] conducted a thorough investigation into the sensitivity of a batch distillation system to several design and operational parameters, including the number of stages, the feed-stage location, and the reflux ratio. Related small-scale thermal separation studies, such as the energy, exergy, economic, and environmental (4E) analysis of solar distillation units equipped with condensers [30] and waste-heat-driven thermal distillation systems [31], likewise emphasize the value of combining energy-based modeling with experiments for decentralized applications.

1.4 Summary of Research Gap

Substantial research has addressed batch distillation modeling, thermodynamics, and experimental validation [32–34]. However, there remains a practical gap in the integration of a low-cost reflux configuration with a modeling approach that can estimate the effective heating power and reflux ratio without direct measurement of internal vapor and liquid flow rates. This limitation is particularly relevant to laboratory and small-scale distillation systems, where flow instrumentation may be unavailable or difficult to implement. In such systems, heat losses, limited instrumentation, and variations in condenser operation can also introduce uncertainty into the estimation of energy input and reflux conditions [35]. Operation at reduced atmospheric pressure further requires the thermodynamic model to reflect the actual operating pressure and corresponding boiling behavior.

This study addresses these issues by developing and experimentally assessing a heat-exchanger-based reflux batch distillation system in which effective heating power is estimated from the measured heat-up period and reflux ratio is inferred from the reflux-condenser duty. These experimentally estimated parameters are then used in an NRTL-based, time-discretized equilibrium-stage model to predict the transient distillate composition and collected volume. The approach is implemented in Python to provide an open and relatively simple modeling framework for small-scale ethanol–water batch distillation.

The specific contributions of this work are as follows: (i) development and fabrication of a four-stage equilibrium representation consisting of a reboiler and three copper bubble-cap trays, coupled with a dual-condenser arrangement comprising a reflux condenser and a main condenser; (ii) an indirect energy-balance method for estimating effective heating power and reflux ratio using temperature, volume, and cooling-water flow measurements without direct vapor or liquid flow metering; (iii) integration of local operating pressure, NRTL vapor–liquid equilibrium, numerical McCabe–Thiele stage analysis, and a time-discretized dynamic solver for

predicting distillate composition and production; (iv) experimental assessment using three batch runs with quantified measurement uncertainty; and (v) comparison with Aspen Plus and a controlled sea-level versus Kathmandu pressure analysis to assess the numerical consistency of the developed framework and the influence of local operating pressure.

The developed framework is intended as a practical modeling and experimental basis for further investigation of small-scale batch distillation systems where detailed flow instrumentation and commercial process-simulation software may not be readily available.

2. MATHEMATICAL MODELING

The performance of the batch distillation unit was evaluated using a framework that included pressure and boiling-point adjustments, VLE modeling, an energy balance to estimate the required power, and reflux ratio determination.

2.1 Atmospheric Pressure and Boiling Point Adjustment

As the experiments were conducted in Kathmandu, Nepal, at an elevation (h) of approximately 1300 m above sea level, the local atmospheric pressure was lower than the standard sea-level pressure ($P_0 = 101325\,Pa$). The local atmospheric pressure for each run was calculated using the isothermal barometric relation:

$$P_{local} = P_0 exp(-\frac{M_{air} g h}{R\,T_{room}}), \quad (1)$$

where $M_{air} = 0.02897\,kg/mol$ is the molar mass of air, $g = 9.81\,m/s^2$ is gravitational acceleration, $R = 8.314\,J/(mol{\cdot}K)$ is the universal gas constant, and T_{room} is the measured ambient room temperature in kelvin. The calculated local pressure was used as the operating pressure for the thermodynamic calculations of each run.

The corresponding boiling temperatures of pure ethanol and water were then estimated using the integrated Clausius–Clapeyron relation:

$$ln(\frac{P_{local}}{P_0}) = -\frac{\Delta H_{vap}}{R}(\frac{1}{T_{b,\,local}} - \frac{1}{T_{b,\,0}}), \quad (2)$$

where $T_{b,0}$ is the normal boiling temperature at standard atmospheric pressure, $T_{b,\,local}$ is the boiling temperature at the calculated local pressure, and ΔH_{vap} is the molar enthalpy of vaporization evaluated at reference ambient temperature. The resulting pure-component boiling temperatures were subsequently used in the mixture bubble-point calculations, while the calculated local pressure was used in the VLE calculations.

2.2 Vapor–Liquid Equilibrium(VLE)

The VLE for the ethanol–water system was described using the modified Raoult's law, with liquid-phase non-ideality captured by the NRTL activity-coefficient model [16]:

$$y_i P = x_i \gamma_i P_i^{sat}(T), \tag{3}$$

where x_i and y_i denote the liquid and vapor phase mole fractions of component i, γ_i is the liquid-phase activity coefficient, P_i^{sat}(T) is the saturation pressure of pure component i at temperature T, and P is the total pressure. Ethanol and water were designated as components 1 and 2, respectively. Saturation pressures were determined using the Antoine equation:

$$log_{10} P_i^{sat} = A_i - \frac{B_i}{T + C_i}, \tag{4}$$

where T is temperature in °C and pressure in mmHg. A_i, B_i, C_i are Antoine constants for component i. The Antoine coefficients were obtained from the DECHEMA Vapor–Liquid Equilibrium Data Collection [36]. For ethanol, two parameter sets were used according to the applicable temperature range: (A = 7.58670), (B = 1281.590), and (C = 193.768) for 78 °C to 203 °C, and (A = 8.11220), (B = 1592.864), and (C = 226.184) for 20 °C to 93 °C. For water, (A = 8.07131), (B = 1730.630), and (C = 233.426) were used. Activity coefficients were computed using the NRTL model. The temperature-dependent binary interaction parameters are given by:

$$\tau_{12} = A_{12} + \frac{B_{12}}{T_K}, \quad \tau_{21} = A_{21} + \frac{B_{21}}{T_K}, \tag{5}$$

where T_K is the absolute temperature. The implementation used $\alpha = 0.30$ with the following interaction parameters for the ethanol–water system: $A_{12} = -0.801$, $B_{12} = 246.2$, $A_{21} = 3.458$, $B_{21} = -586.1$. These parameters correspond to the standard NRTL set reported by Guevara Luna et al. [10], who also evaluated the model against sub-atmospheric VLE data. For the binary mixture, the NRTL activity coefficients were calculated using:

$$G_{ij} = exp\left(-\alpha \tau_{ij}\right), \tag{6}$$

with

$$\ln \gamma_1 = x_2^2 \left[\tau_{21}\left(\frac{G_{21}}{x_1 + x_2 G_{21}}\right)^2 + \frac{\tau_{12} G_{12}}{\left(x_2 + x_1 G_{12}\right)^2}\right], \ \ln \gamma_2 = x_1^2 \left[\tau_{12}\left(\frac{G_{12}}{x_2 + x_1 G_{12}}\right)^2 + \frac{\tau_{21} G_{21}}{\left(x_1 + x_2 G_{21}\right)^2}\right]. \tag{7}$$

For each specified liquid composition, the bubble-point temperature was obtained by numerically solving the total-pressure equation:

$$x_1 \gamma_1 P_1^{sat}(T) + x_2 \gamma_2 P_2^{sat}(T) = P_{local}, \tag{8}$$

where P_{local} is the local atmospheric pressure determined in Section 2.1. The resulting temperature is thus the bubble-point temperature at the given liquid composition and local operating pressure. Once the bubble-point temperature was found, the equilibrium vapor-phase ethanol mole fraction was computed as:

$$y_1 = \frac{x_1 \gamma_1 P_1^{sat}(T)}{x_1 \gamma_1 P_1^{sat}(T) + x_2 \gamma_2 P_2^{sat}(T)}. \tag{9}$$

This procedure was repeated across the entire liquid composition range to generate the isobaric ethanol–water equilibrium curve, which was subsequently used in the McCabe–Thiele analysis and the dynamic simulation.

2.3 Energy Balance and Power Input

The actual heat transferred from the liquefied petroleum gas (LPG) burner to the distillation system was not measured directly due to heat losses to the surroundings. Instead, an effective heating power was determined from the experimentally measured time required for the initial feed to reach its bubble point. The calculation accounted for the sensible heat required to heat the ethanol–water feed and to raise the temperature of the boiler pot.

The thermophysical properties required for the energy calculations were evaluated from literature-based correlations and tabulated data. The specific heat capacities of pure water and ethanol were calculated using the correlations reported by Osborne et al. [37] and Miyazawa et al. [38], respectively. The specific heat capacity of the ethanol–water mixture was obtained by combining the mole-fraction-weighted pure-component contributions with an excess heat-capacity term based on the formulation of Gaulhofer et al. [39]:

$$C_{p,mix} = \frac{x_e C_{p,e} M_e + x_w C_{p,w} M_w + C_p^E}{x_e M_e + x_w M_w}, \tag{10}$$

where x_e and x_w are the liquid-phase mole fractions of ethanol and water, M_e and M_w are their molar masses, and C_p^E is the molar excess heat-capacity contribution. The mixture density and thermophysical constants were obtained from the tabulated ethanol–water data of Washburn [40] and Lange's Handbook [41] using two-dimensional linear interpolation with respect to ethanol mass fraction and temperature. The density was used to convert liquid volume to mass. For vaporization calculations, saturated liquid and vapor enthalpy data for ethanol and water were obtained from Perry's Chemical Engineers' Handbook [1] and interpolated using cubic splines. The mixture latent heat was then evaluated from the difference between the composition-weighted vapor and liquid molar enthalpies:

$$\Delta H_{vap,mix} = H_V - H_L. \tag{11}$$

These property correlations were evaluated at the relevant temperature and composition conditions during the calculations. The sensible heat absorbed by the feed was calculated by numerically integrating the temperature-dependent specific heat capacity of the initial mixture:

$$Q_s = \int_{T_0}^{T_{b,0}} M_{feed} C_{p,mix}\left(T, x_{e,0}\right) dT, \tag{12}$$

where M_{feed} is the initial feed mass, $x_{e,0}$ is the initial liquid-phase ethanol mole fraction, $C_{p,mix}$ is the specific heat capacity of the ethanol–water mixture, T_0 is the measured initial feed temperature, and $T_{b,0}$ is the bubble-point temperature of the initial feed at the local operating pressure. Then the heat absorbed by the boiler pot was calculated as:

$$Q_{pot} = m_{pot} C_{p,pot}\left(T_{b,0} - T_0\right), \tag{13}$$

where m_{pot} and $C_{p,pot}$ are the mass and specific heat capacity of the boiler pot, respectively. In the present study, m_{pot} = 6.056 kg and $C_{p,pot}$ = 511 J/(kg·K). The effective heating power was then obtained from the total heat required during the measured heat-up period:

$$P_{supplied} = \frac{Q_s + Q_{pot}}{t_{bpt}}, \tag{14}$$

where t_{bpt} is the measured time from the initial feed condition to the onset of boiling, identified by the first appearance of vapor and the corresponding thermometer indication. The mixture heat capacity was evaluated at the initial liquid composition, while its temperature dependence was accounted for over the heat-up interval. The resulting $P_{supplied}$ represents the effective average heating power used in the subsequent dynamic simulation. This power was assumed to remain constant throughout the distillation.

2.4 Reflux Ratio

The reflux ratio was determined using the heat duty of the reflux condenser and the effective heating power measured in the heat-up experiment. The reflux-condenser heat duty was calculated from the cooling-water flow rate and temperature rise:

$$Q_{cond} = \dot{m}_{cw} C_{p,cw} \left(T_{out} - T_{in} \right), \tag{15}$$

where $\dot{m}_{cw}$ is the cooling-water mass flow rate, $C_{p,cw}$ is the specific heat capacity of water, and T_{in} and T_{out} are the inlet and outlet cooling-water temperatures, respectively. The water density and specific heat capacity were evaluated at the mean cooling-water temperature.

In determining the reflux heat duty, the condensed column vapor was assumed to contribute only its latent heat of condensation; sensible heat associated with cooling the vapor or condensate was neglected. Thus, the measured condenser duty was attributed to the latent heat released by the reflux stream. The reflux ratio was then calculated from the condenser duty relative to the remaining effective heat input available for distillate generation:

$$R = \frac{Q_{cond}}{P_{supplied} - Q_{cond}}, \tag{16}$$

where $P_{supplied}$ is the effective heating power determined from Equation (14). The resulting reflux ratio was treated as constant throughout the dynamic simulation.

2.5 McCabe–Thiele Stage Analysis

The relationship between the instantaneous bottoms composition and distillate composition was obtained using a numerical McCabe–Thiele stage-stepping procedure. The ethanol–water equilibrium curve generated from the NRTL model in Section 2.2 was first discretized over the full liquid-composition range. For a specified distillate composition and reflux ratio R, the rectifying operating line was defined as:

$$y = \frac{R}{R+1} x + \frac{x_D}{R+1}, \tag{17}$$

where x and y are the liquid- and vapor-phase ethanol mole fractions, respectively.

Starting from $x = x_D$, the operating line and equilibrium relation were alternately evaluated for the specified number of theoretical stages. At each stage, the vapor composition was first obtained from the operating line, and the corresponding liquid composition was determined by interpolation of the pre-calculated equilibrium curve. After the prescribed number of stages, the resulting liquid composition was taken as the bottoms composition x_W. Thus, the stage calculation generated a numerical relationship between x_D and x_W is:

$$x_W = F(x_D). \tag{18}$$

The calculation was repeated over 400 distillate-composition points, with increased point density toward the high-distillate-composition region to resolve the more rapid variation of x_D with x_W. The resulting data were sorted, and duplicate values were removed before a cubic-spline interpolation was constructed:

$$x_D = S(x_W). \tag{19}$$

This spline provided the instantaneous distillate composition directly from the current bottoms composition during the dynamic simulation, avoiding repeated stage-wise calculations at every time step. The column was represented by four theoretical equilibrium stages, consisting of the reboiler and three bubble-cap trays. Each stage was treated as an ideal equilibrium stage, and no explicit Murphree tray-efficiency correction was applied. The resulting stage-composition relationship was subsequently coupled with the transient material and energy balances.

2.6 Time-Discretized Dynamic Simulation

The dynamic behavior of the batch column after the onset of boiling was predicted using an explicit time-marching equilibrium-stage model implemented in Python. The numerical procedure consists of a one-time pre-processing stage, followed by a fixed-step simulation loop. The overall algorithm is summarized in Figure 1.

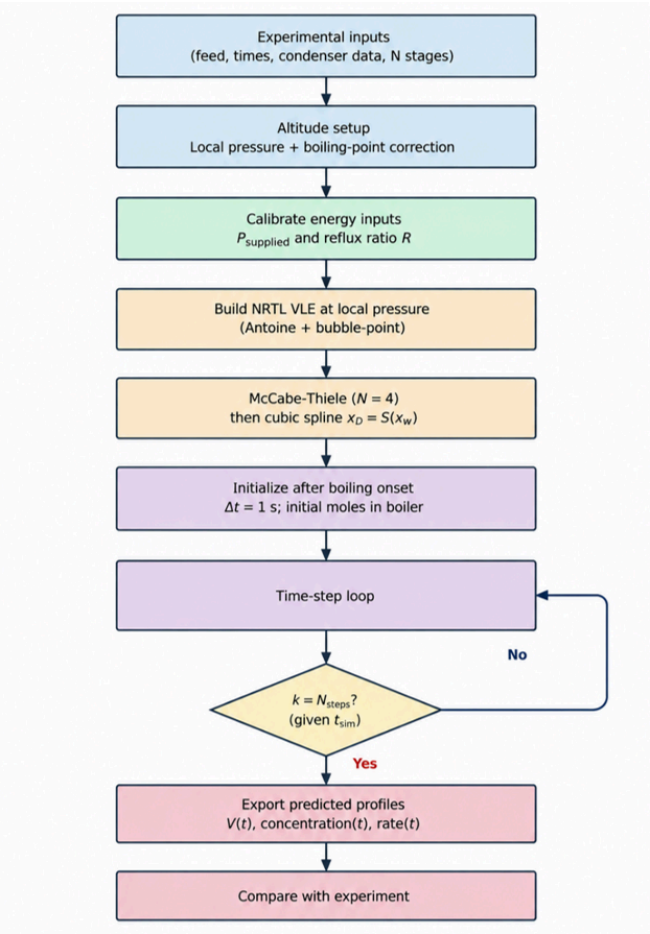


Figure 1. Flowchart of the NRTL batch-distillation solver: altitude-corrected VLE and energy estimation, McCabe–Thiele spline construction, and the 1-second time-stepping loop used to predict distillate volume and composition.

The equilibrium relation, estimated heating power, and reflux ratio were used as fixed model inputs. The simulation was performed with a time step of $\Delta t = 1$ second. The simulated separation period was defined as:

$$t_{sim} = t_{total} - t_{bpt}, \tag{20}$$

where t_{total} is the total experimental duration and t_{bpt} is the measured time to the onset of boiling.

The measured initial feed volume and ethanol concentration (v/v%) were converted to the initial ethanol and water inventories required by the model. The ethanol volume was calculated directly from the initial concentration and total feed volume. In contrast, the water volume was determined iteratively to match the measured initial total feed volume (Table 1), explicitly accounting for non-ideal liquid volumetric contraction through the mixture-density correlation at the initial feed temperature. The resulting component masses were converted to moles to obtain the initial liquid ethanol mole fraction used for initialization.

At each time step, the ethanol mole fraction of the remaining boiler liquid was calculated from the component inventories. The corresponding bubble-point temperature was obtained from the NRTL model, and the instantaneous distillate composition was determined from the four-stage $x_W - x_D$ relationship developed in Section 2.5. The first-stage equilibrium vapor composition was used to evaluate the latent heat of the current mixture. The molar distillate withdrawal was then calculated from the energy balance:

$$\Delta n_D = \frac{P_{supplied}\,\Delta t}{(1+R)\Delta H_{vap}}. \tag{21}$$

The withdrawn ethanol and water were calculated from the instantaneous distillate composition, and component material balances were used to update the remaining boiler inventories. The collected component amounts were subsequently converted to mass and mixture volume using the corresponding mixture-density correlation [40]. Cumulative and instantaneous ethanol concentrations were then calculated for comparison with the experimental measurements. The calculation was continued until the experimental simulation time was reached or the remaining boiler inventory fell below the specified minimum number of moles.

3. EXPERIMENTAL ASSESSMENT

This section describes the experimental framework used to assess the performance of the altitude-corrected NRTL-based dynamic model. The apparatus design and fabrication are summarized only to the extent necessary for reproducibility, while emphasis is placed on instrumentation, measurement procedures, estimation of the effective heat input and reflux ratio, and the experimental data used to assess model performance.

3.1 Apparatus Overview

Figure 2 shows a schematic diagram of the batch distillation unit and the process flow. The distillation setup includes five major components: the boiler (or reboiler), the distillation column with bubble-cap trays, the reflux condenser, the main condenser, and the distillate collection system. When batch distillation begins, the feed material is placed in the boiler, where it is heated to produce a mixture of ethanol and water vapor. The vapor rises through the column and passes over several bubble-cap trays that enhance vapor–liquid contact and promote separation. Part of the vapor is then taken to the reflux condenser, where a portion of the ethanol and water is condensed and recycled back into the column to achieve a higher purity level of ethanol; the remaining portion is sent to the main condenser, where the vapor is completely condensed into distillate, which is collected and later measured to determine its volume and concentration.

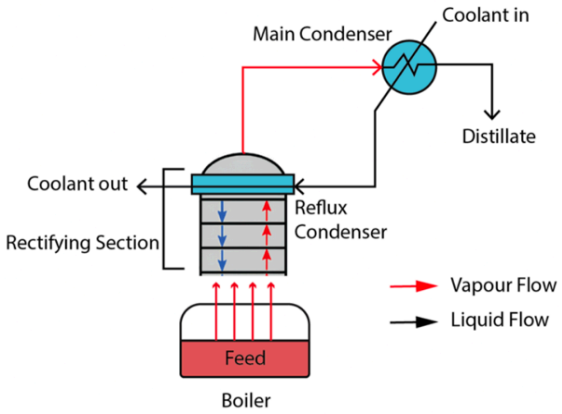


Figure 2. Schematic representation of the batch distillation apparatus, illustrating vapor flow from the boiler through the rectifying section and condensation in the reflux and main condensers, showing the resulting distillate and reflux streams.

3.2 Design and Fabrication

The boiler was fabricated from stainless steel, with an approximate capacity of 29 L, and heated by an LPG burner. Based on the McCabe–Thiele method, the distillation column was designed with four theoretical stages: three bubble-cap trays and the reboiler. Double-pipe water-cooled condensers were used for vapor condensation. Bubble-cap trays provided vapor–liquid contact, while silicone gaskets and bolted flanges provided sealing and allowed disassembly. The cooling-water flow to the reflux condenser was adjustable. The CAD models of the main components and their assembly are shown in Figure 3.

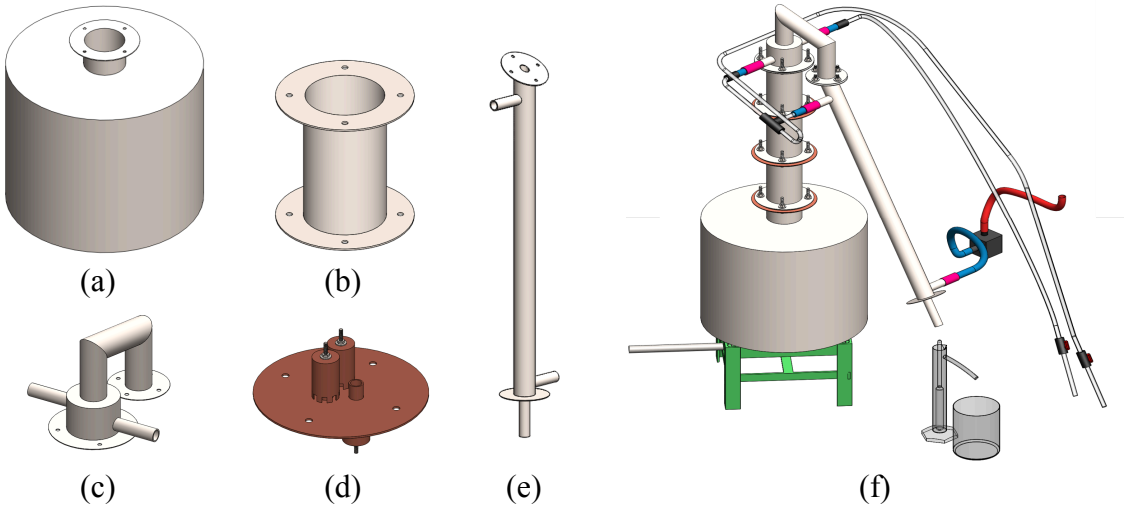


Figure 3. Computer-Aided Design (CAD) models of the components of the experimental setup: (a) Boiler, (b) Column unit, (c) Reflux condenser, (d) Bubble cap tray, (e) Main condenser, (f) Full assembly.

The unit was fabricated locally using stainless steel and copper through conventional workshop processes. The components were assembled using bolted connections and silicone gaskets to provide sealed joints and facilitate disassembly. Figure 4 shows the assembled prototype.

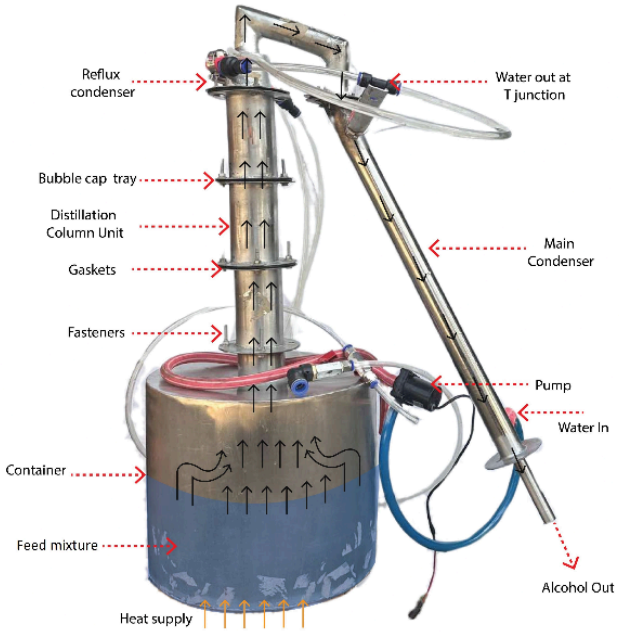


Figure 4. Fabricated prototype of the developed batch distillation apparatus for ethanol–water separation.

3.3 Instrumentation and Measurement

Thermometers were placed at the boiler, column top, and condenser cooling-water inlet/outlet. Distillate volume was measured with Class B graduated cylinders, and ethanol content was measured with an alcoholometer. Cooling-water flow was obtained by a volumetric timing method. These measurements provided the estimated energy inputs (heating power, condenser duty, reflux ratio) of Sections 2.3 and 2.4 and the evaluated outputs (composition and volume histories) compared in Section 4.

Cooling water flow rates were calculated using a simple volumetric method, in which the time required to fill a certain volume was measured. For each experimental run, the time required to reach boiling and the duration until distillate collection stopped (as the volume of the bottom mixture decreased) were recorded. During each run, the cumulative distillate volume and its concentration were logged, along with the heating time, the cooling-water inlet and outlet temperatures, and the cooling-water flow rate; all instrument least counts are summarized in the uncertainty analysis in Section 4.7.

3.4 Experimental Setup and Operating Conditions

The apparatus was operated to test the developed model in three experimental runs with different feed volumes and concentrations. Run 1 used distillate recovered from fermented mash,

conditioned and diluted to an initial ethanol concentration of 20% (v/v). Because the original mash may contain residual impurities, its exact composition is uncertain. Run 2 was fed with distillate recovered from Run 1, diluted to 27% (v/v). Run 3 used distillate recovered from Run 2 together with additional ethanol from intermediate test runs, diluted to 30% (v/v). All experiments were carried out under ambient conditions in Kathmandu, and the altitude-induced pressure reduction was accounted for in all calculations. Table 1 lists the initial feed, operating, and environmental conditions for the three runs. Because operating parameters, particularly cooling-water flow rates and condenser temperatures, exhibited natural fluctuations during batch operation, their time-averaged values were used as constant boundary inputs for the dynamic simulations. These measured values were used directly, without adjustment, as inputs to the model.

Table 1. Initial feed, operating, environmental, and calculated composition conditions for the experimental runs.

Category	Parameters	Run 1	Run 2	Run 3
Initial feed conditions	Initial feed volume (L)	4.0	3.0	3.5
	Initial ethanol concentration (v/v%)	20	27	30
	Initial feed temperature (°C)	18.3	15.1	17.0
Operating conditions	Total experiment duration (min)	40	38	53
	Time to reach boiling point (min)	12.00	14.08	15.67
	Ambient temperature (°C)	19.1	17.0	18.0
Reflux condenser water	Inlet temperature (°C)	25.0	20.7	24.73
	Outlet temperature (°C)	29.1	31.95	37.1
	Flow rate (mL/s)	36.00	11.69	9.71
Initial amount in moles	Ethanol (mol)	13.732	13.951	18.048
	Water (mol)	181.162	125.495	141.065
	Total (mol)	194.894	139.446	159.113
	Ethanol mole fraction	0.0705	0.1000	0.1134

At the start of each trial, the initial temperature was recorded, and heating was initiated using an LPG burner. The time required for the mixture to reach the boiling point was recorded. Once vapor condensation began, distillate samples were collected periodically. A graduated cylinder was used to collect each sample up to a certain volume, and the corresponding collection time was recorded. The ethanol concentration of each sample was then measured using an alcoholometer. The inlet and outlet cooling-water temperatures and cooling-water flow rate were also recorded for each sample. Where multiple readings were taken, their average value was used. The measurements for each run are summarized in Table 1. and were used as inputs to the model described in Section 2.

4. RESULTS AND DISCUSSION

4.1 Local Operating Conditions and Pure-Component Boiling Points

The local atmospheric pressure and corresponding pure-component boiling temperatures were calculated separately for each experimental run using the procedure described in Section 2.1. The results are presented in Table 2.

Table 2. Run-specific local atmospheric pressure and pure-component boiling points.

Parameter	Run 1	Run 2	Run 3
Local atmospheric pressure (Pa)	87032.3	86936.6	86982.4
Ethanol boiling point(°C)	74.68	74.66	74.67
Water boiling point(°C)	96.06	96.04	96.05

The calculated local atmospheric pressure varied by approximately 0.11% across the three runs, while the calculated boiling temperatures of pure ethanol and water varied by at most 0.02 °C. Thus, the run-to-run variation in atmospheric conditions had a negligible effect on the calculated pure-component boiling points. The run-specific values were nevertheless retained for consistency with the calculations for each experimental run.

4.2 Energy Balance and Reflux Ratio

The calculated effective heating power, reflux-condenser duty, and reflux ratio for the three experimental runs are summarized in Table 3. The effective heating power decreased from Run 1 to Run 3, while the reflux ratio varied with the measured condenser duty and the available heating power. These experimentally derived values were used as fixed inputs in the dynamic simulations.

Table 3. Energy-balance results for the experimental runs, including supplied heating power, condenser heat duty, and reflux ratio.

Parameter	Run 1	Run 2	Run 3
Supplied heating power (W)	1580.46	1061.23	1025.34
Condenser heat duty (W)	614.65	547.71	499.53
Reflux ratio	0.636	1.067	0.950

4.3 VLE and McCabe–Thiele Analysis: Run 3

The calculated ethanol–water equilibrium behavior at the local atmospheric pressure in Kathmandu is presented in Figure 5. The equilibrium curve shows the expected enrichment of ethanol in the vapor phase compared with the liquid phase over the composition range considered. For the initial feed composition of Run 3, the calculated boiling temperature was 81.84 °C. At the reflux ratio of Run 3, the four-stage McCabe–Thiele construction predicts progressive enrichment of ethanol from the bottoms to the distillate. The resulting relationship between the instantaneous bottoms and distillate ethanol mole fractions is shown in Figure 6. The relationship was represented using a cubic spline to provide the corresponding distillate composition as the bottoms composition changed during the batch operation.

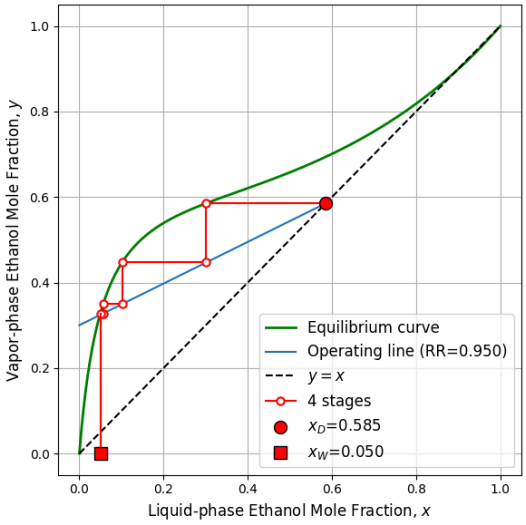


Figure 5. McCabe–Thiele diagram for Run 3 illustrating the equilibrium curve, operating line, and four theoretical stages (reflux ratio = 0.950)

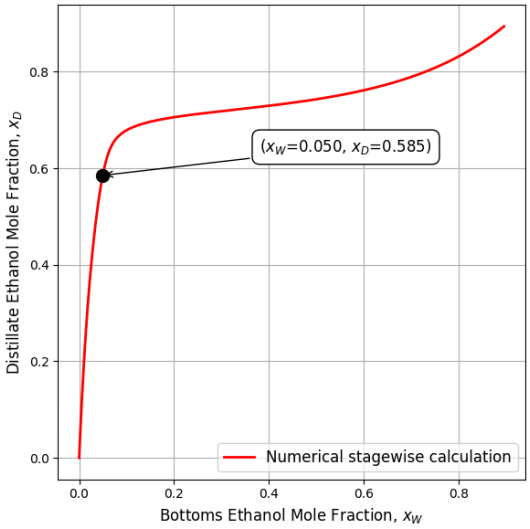


Figure 6. Relationship between bottoms ethanol mole fraction and distillate ethanol mole fraction for Run 3, obtained from the four-stage McCabe–Thiele stage-stepping calculation at a reflux ratio of 0.950.

Figure 6 shows that the distillate ethanol mole fraction decreases as the bottoms ethanol mole fraction decreases. For example, at $x_W = 0.050$, the corresponding distillate composition is approximately $x_D = 0.585$. The decrease in x_D with decreasing x_W indicates the gradual reduction in distillate enrichment as ethanol is depleted from the boiling liquid. At low x_W, the relationship becomes steeper, showing that the distillate composition becomes increasingly sensitive to changes in the remaining ethanol concentration in the bottoms.

4.4 Dynamic Distillation Profiles: Run 3

The dynamic distillation results for Run 3 are presented in Figures 7–10. The model used the estimated heating power of 1025.3 W and reflux ratio of 0.950. Time in the plots is referenced to the start of heating, with boiling beginning at 15.67 min. The model and experimental results are compared in terms of cumulative distillate volume, cumulative ethanol concentration, interval-average ethanol concentration, and distillate collection rate.

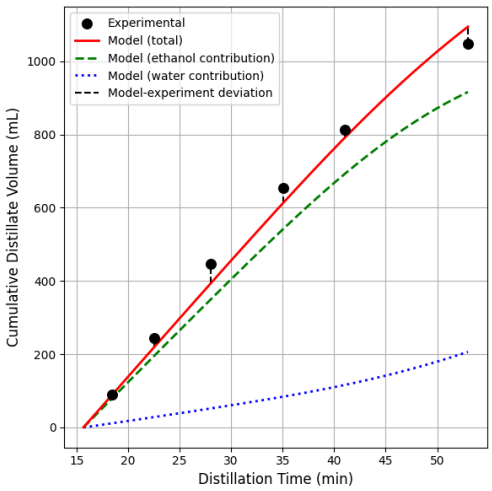


Figure 7. Cumulative distillate volume versus distillation time for Run 3. Experimental cumulative distillate volumes are compared with model predictions; dashed vertical lines indicate the model–experiment deviation.

The cumulative distillate volume increased throughout the distillation period, as expected. The model followed the overall trend of the experimental data, although a difference between the predicted and measured cumulative volumes was observed.

Table 4. Comparison of experimental and model-predicted distillate ethanol concentrations of Run 3 at different distillation times.

Distillation Time (min)	Experimental (v/v%)	Model (v/v%)	Deviation (%)
18.48	87	89.22	2.49
22.5	86	89.10	3.48
28	85	88.87	4.35
35.05	84	88.38	4.96
41	83	87.53	5.18
53	79	83.70	5.62

The cumulative ethanol concentration decreased with increasing distillation time (Table 4 and Figure 8), reflecting the progressive depletion of ethanol from the boiling liquid. The experimental concentration decreased from 87% to 79% (v/v), while the model predicted a decrease from 89.22% to 83.70% (v/v). Despite the difference in magnitude, both results show the same overall decreasing trend. All percentage deviations are evaluated relative to the model predictions, calculated as (|Experimental − Model| / Model) × 100%.

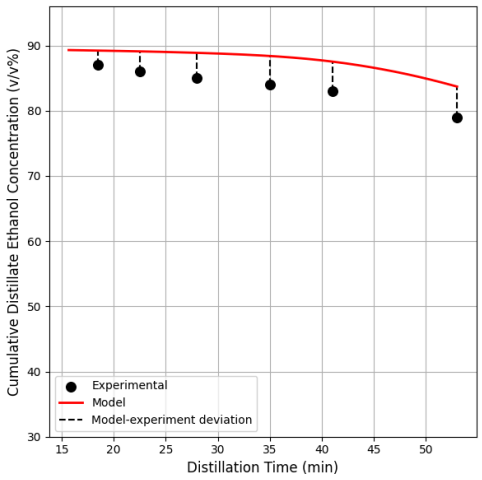


Figure 8. Cumulative distillate ethanol concentration versus distillation time for Run 3. Experimental cumulative concentrations are compared with the model prediction; dashed vertical lines indicate the model–experiment deviation.

The variation in distillate ethanol concentration during Run 3 is shown in Figure 9. The experimental measurements are presented at the corresponding measurement intervals, together with the model predictions over the same distillation period.

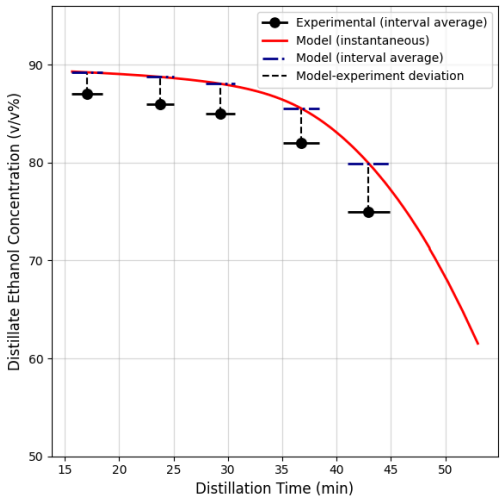


Figure 9. Comparison of experimental and model-predicted distillate ethanol concentration with distillation time for Run 3. Experimental interval-average values, instantaneous model predictions, interval-average model predictions, and model–experiment deviations are shown.

The experimental ethanol concentrations represent interval-average measurements rather than instantaneous values. Therefore, the model predictions were also averaged over the corresponding time intervals to provide a direct comparison with the experimental data. The interval-average model results follow the experimental trend, with the model generally predicting a higher ethanol concentration throughout the distillation period.

The variation in distillate collection rate during Run 3 is shown in Figure 10. The experimental rates are interval-average values and are compared with the corresponding interval-average model predictions.

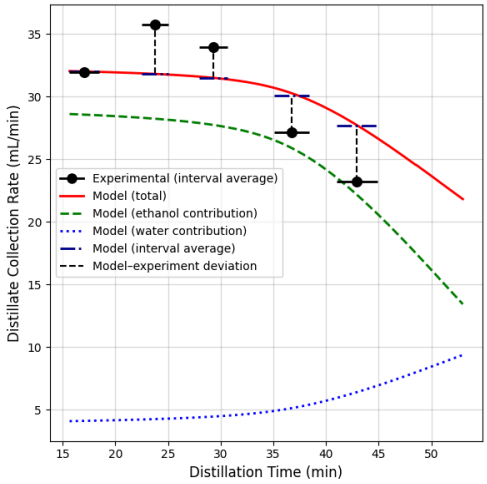


Figure 10. Distillate collection rate versus distillation time for Run 3. Experimental interval-average rates are compared with model-predicted total, ethanol, water, and interval-average rates; dashed vertical lines indicate the model–experiment deviation.

The initial experimental collection rate was close to the model prediction. However, during subsequent intervals, the experimental rate exceeded the model prediction, particularly around 25 minutes. This resulted in a higher cumulative distillate volume than the model predicted over this period. The higher collection rate was likely influenced by fluctuations in the reflux-condenser cooling-water flow, which caused the actual reflux conditions to vary rather than remain at the constant reflux ratio of 0.950 used in the simulation. This variation represents a limitation of the experimental setup and is a possible source of deviation between the experimental and model results. As the cooling-water flow was maintained more consistently, the experimental collection rate decreased and fell below the model prediction in the later intervals. Consequently, the difference in cumulative volume was reduced as the experimental result approached the model prediction.

4.5 Experimental Testing Across Runs

Table 5 summarizes the final distillate ethanol concentrations and total collected volumes obtained experimentally and predicted by the NRTL model for Runs 1–3, where percentage deviations are evaluated relative to the model predictions.

Table 5. Comparison of the model against experimental results for Runs 1–3 in terms of final distillate ethanol concentration and collected volume.

Parameter	Source	Run 1	Run 2*	Run 3
Ethanol concentration (v/v%)	Model	64.75	86.13	83.70
	Experimental	58	84	79
	Deviation (%)	10.42%	2.47%	5.62%

Distillate volume (mL)	Model	1170.77	711.80	1094.49
	Experimental	1182	593	1049
	Deviation (%)	0.96%	16.69%	4.16%

** Run 2 was affected by column flange gasket leakage during operation.*

The model–experiment deviations in final ethanol concentration were 10.42%, 2.47%, and 5.62% for Runs 1–3, respectively, while the corresponding deviations in total collected distillate volume were 0.96%, 16.69%, and 4.16%. The differences among the runs were also influenced by their respective experimental conditions. Run 1 used fermented mash as the feed, which exhibited irregular boiling behavior and contained residual non-volatile components. Run 2 was affected by leakage at the column seal/flange near the first (bottom) bubble-cap tray, resulting in liquid loss during operation and a larger deviation in collected distillate volume. Run 3 was conducted with the prepared feed under sound mechanical sealing and more controlled operating conditions, resulting in relatively low deviations in both final concentration (5.62%) and collected volume (4.16%). Accordingly, Run 3 provides the most controlled experimental case for assessing the model performance. Across all three runs, the concentration root-mean-square error (RMSE) was 4.91 v/v% (normalized RMSE = 6.66%), while the volume RMSE was 73.73 mL (NRMSE = 7.83%).

4.6 Effect of Altitude Correction

The effect of operating pressure on the model predictions was evaluated by comparing results at standard atmospheric pressure with those at the corresponding Kathmandu operating pressure, while keeping the estimated heating power and reflux ratio constant. The results are presented in Table 6.

Table 6. Comparison of sea-level and Kathmandu-pressure predictions at fixed heating power and reflux ratio.

Parameter	Pressure condition	Run 1	Run 2	Run 3
Distillate ethanol mole fraction	Sea level	0.340	0.622	0.578
	Kathmandu	0.341	0.623	0.579
	Deviation (%)	0.29%	0.16%	0.21%
Total distillate (mol)	Sea level	38.351	17.015	27.282
	Kathmandu	38.193	16.964	27.192
	Deviation (%)	0.41%	0.30%	0.33%

The pressure correction resulted in small changes in both predicted distillate composition and total distillate amount. The relative difference in distillate ethanol mole fraction ranged from 0.16% to 0.29%, while that in total distillate amount ranged from 0.30% to 0.41% across the three runs. These results indicate that the reduced atmospheric pressure had a small effect on the predicted separation under the fixed heating-power and reflux-ratio conditions considered. The Kathmandu operating pressure was therefore retained in the final model for consistency with the experimental operating conditions.

4.7 Measurement Uncertainty Analysis

Measurement uncertainties were assessed to quantify how instrument resolution affects derived operational variables and model predictions. The analysis was conducted within the framework of JCGM 101 (GUM Supplement 1) [42], employing Monte Carlo propagation of uncertainty with rectangular probability distributions assigned to the instrument-resolution inputs. The Monte Carlo approach was chosen because multiple measurement uncertainties propagate through both the energy-balance calculations and the nonlinear dynamic distillation model, rendering direct analytical uncertainty propagation impractical.

For an instrument with least count Δ, the standard uncertainty was computed as, $u = \Delta/\sqrt{12}$, consistent with a rectangular distribution spanning $\pm\Delta/2$. The least counts for model inputs were 0.1° C for temperature measurements, 1 second for timing, 5 mL for initial feed and cooling-water volumes, and 1 v/v% for initial alcoholometer readings. A 1000-sample Monte Carlo simulation for Run 3 propagated these input uncertainties through the energy-balance calculations and the dynamic four-stage distillation model. The resulting standard uncertainties are listed in Table 7.

Table 7. Monte Carlo-propagated standard uncertainties of derived operational variables and model predictions for Run 3 (N = 1000).

Parameter / Variable	Nominal Value	Propagated Standard Uncertainty	Relative Uncertainty (%)
Effective heating power	1025.34 W	±1.94 W	0.19
Condenser heat duty	499.53 W	±2.46 W	0.49
Reflux ratio	0.950	±0.010	1.05
Final distillate concentration	83.70 v/v%	±0.34 v/v%	0.41
Total distillate volume	1094.49 mL	±3.40 mL	0.31

The Monte Carlo uncertainty analysis shows that instrument-resolution limits contribute only minimal uncertainty to the derived operational variables (relative uncertainties ≤ 1.05%) and to the final model predictions (relative uncertainties ≤ 0.41%). These propagated uncertainties are smaller than the model–experiment deviations found in Run 3, which were 5.62% for concentration and 4.16% for total distillate volume. Thus, the observed deviations cannot be attributed solely to the specified instrument-resolution uncertainties and may also arise from model assumptions and other experimental effects.

4.8 Sensitivity Analysis of Model Predictions

The sensitivity of the predicted batch-distillation performance to reflux ratio, number of theoretical stages, and heating power was evaluated using a one-factor-at-a-time approach [43], using Run 3 as the baseline. For each analysis, the selected parameter was varied while the other model inputs were kept at their Run 3 values. The reflux ratio was varied from 0.76 to 1.14, the number of theoretical stages from 2 to 6, and the heating power from 820.27 to 1230.41 W. The effects on ethanol mole fraction and total distillate production are presented in Figure 11.

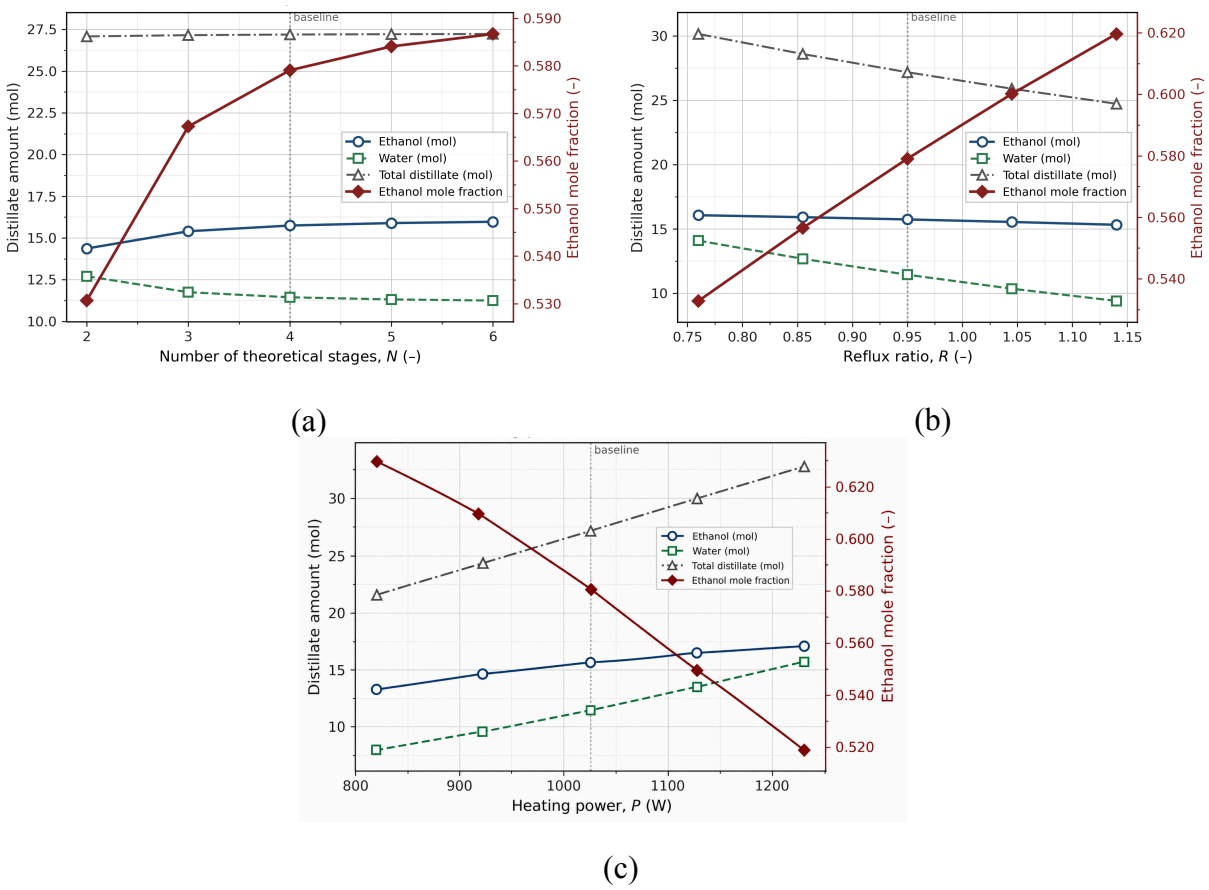


Figure 11. Sensitivity of the predicted batch-distillation performance to (a) reflux ratio, (b) number of theoretical stages, and (c) heating power, showing their effects on distillate ethanol mole fraction and total distillate production.

As shown in Figure 11(a), increasing the reflux ratio from 0.76 to 1.14 increased the predicted ethanol mole fraction from 0.5328 to 0.6196, while the total distillate amount decreased from 30.168 to 24.741 mol. Increasing the number of theoretical stages from 2 to 6 increased the ethanol mole fraction from 0.5307 to 0.5867, while the total distillate amount changed only slightly from 27.079 to 27.224 mol [Figure 11(b)]. The increase in ethanol enrichment became progressively smaller beyond four stages. In contrast, increasing the heating power from 820.27 to 1230.41 W increased the total distillate amount from 21.641 to 32.826 mol but decreased the ethanol mole fraction from 0.6280 to 0.5193 [Figure 11(c)]. These results show that reflux ratio and heating power have opposing effects on distillate purity and production, whereas increasing the number of stages beyond four provides comparatively limited additional enrichment under the Run 3 operating conditions.

4.9 Numerical Benchmarking Using Aspen Plus

The experimental batch distillation system setup was replicated in Aspen Plus using the BatchSep batch-distillation model [44]. The NRTL property method was used to describe the vapor–liquid equilibrium of the ethanol–water system, with the operating pressure set to the atmospheric pressure corresponding to Kathmandu. The feed composition, initial amount, initial

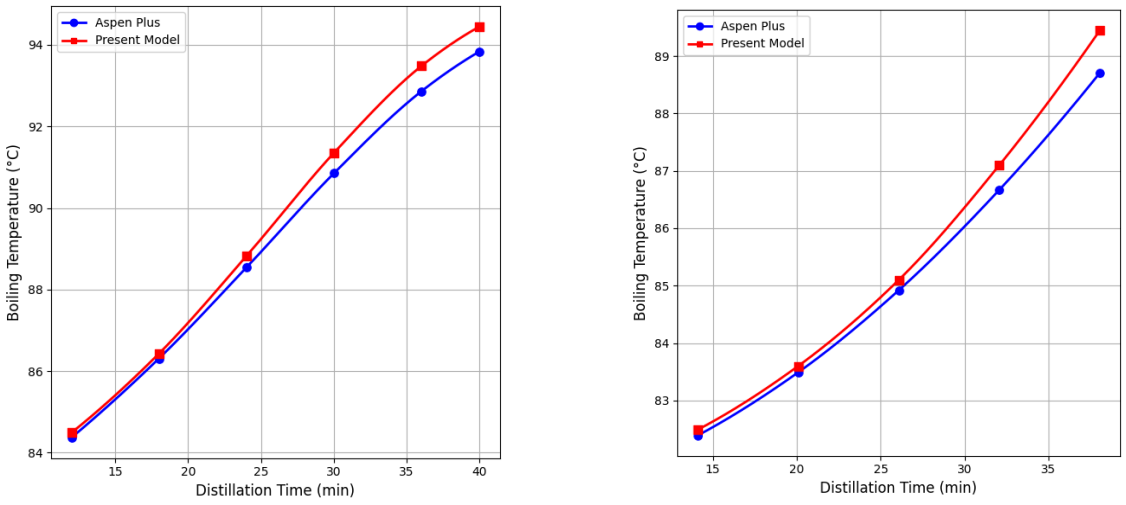
temperature, reflux ratio, and other operating conditions were specified according to the three experimental runs. The column was represented using equilibrium stages corresponding to the experimental column with three bubble-cap trays. The present model was compared with the Aspen results in terms of distillate ethanol mole fraction, total distillate amount, and ethanol and water in the distillate. The relative deviation was calculated with respect to the corresponding Aspen result.

Table 8. Numerical benchmarking of the present model against Aspen Plus simulation results for the three experimental runs.

Parameter	Source	Run 1	Run 2	Run 3
Ethanol mole fraction	Aspen Plus	0.351	0.613	0.5763
	Present model	0.341	0.623	0.5791
	Deviation (%)	2.91%	1.57%	0.47%
Total distillate (mol)	Aspen Plus	36.170	16.312	26.218
	Present model	38.193	16.964	27.192
	Deviation (%)	5.59%	4.00%	3.71%
Ethanol in distillate (mol)	Aspen Plus	12.693	9.998	15.110
	Present model	13.013	10.560	15.746
	Deviation (%)	2.52%	5.62%	4.21%
Water in distillate (mol)	Aspen Plus	23.477	6.314	11.108
	Present model	25.180	6.404	11.446
	Deviation (%)	7.25%	1.42%	3.04%

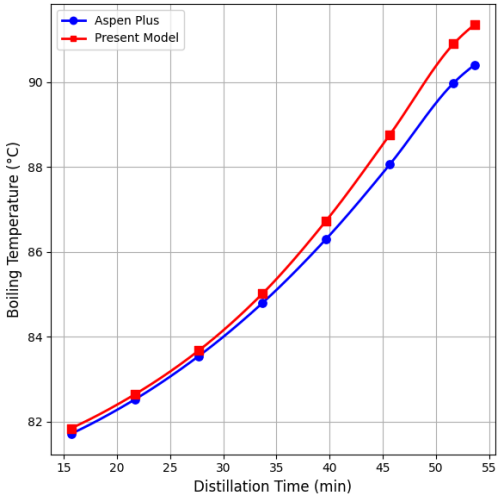
As shown in Table 8, the ethanol mole fractions predicted by the present model agreed closely with the Aspen Plus results, with relative deviations of 2.91%, 1.57%, and 0.47% for Runs 1–3, respectively. The corresponding deviations in total distillate amount were 5.59%, 4.00%, and 3.71%. Overall, the relative deviations between the present model and Aspen Plus were below 7.3% for all quantities considered. The close agreement indicates that the developed equilibrium-stage formulation produces results consistent with an established process-simulation framework under the investigated operating conditions. The Aspen Plus comparison therefore provides a numerical benchmark and software consistency check for the custom Python code, complementing the experimental evaluation presented in the preceding sections.

The transient temperature profiles predicted by the present model and Aspen Plus for the three experimental runs are shown in Figure 12. The comparison was performed using the same operating conditions specified for each run. The predicted temperature profiles exhibit similar overall trends, with the temperature increasing during the heating period and subsequently following the changing boiling conditions during distillation.



(a)

(b)



(c)

Figure 12. Comparison of transient temperature profiles predicted by the present model and Aspen Plus as a function of distillation time for (a) Run 1, (b) Run 2, and (c) Run 3.

As shown in Figure 12, the present model closely follows the Aspen Plus predictions for all three runs. Run 1 shows close agreement throughout the distillation period, with only small differences at times. Similar behavior is observed for Run 2, although slightly larger differences occur at certain points. For Run 3, the two profiles remain closely aligned over the investigated period. The overall agreement in the transient temperature profiles provides an additional numerical check of the developed model's dynamic behavior and indicates consistency with Aspen Plus predictions under the investigated operating conditions.

4.10 Discussion

4.10.1 Model Performance and Experimental Evaluation

The experimental results demonstrate reasonable agreement with the model, yielding predictive deviations that align with the ranges reported for validated batch-distillation models [7, 20, 45]. Run 3 provided the most reliable baseline for this assessment, as Run 1 experienced irregular boiling from the fermented mash and Run 2 was distinctly leakage-affected due to flange gasket seepage. Across the experiments, the model predicted distillate composition with lower relative deviation than total collected volume. This discrepancy aligns with the physical mechanics of the system: composition is primarily governed by VLE, whereas production volume is highly sensitive to heating power and reflux conditions. While the model assumed a constant effective heat input and reflux ratio, actual experimental conditions naturally fluctuated, as evidenced by the variable collection rate and cooling-water flow in Run 3. Furthermore, remaining minor discrepancies can be attributed to deliberate model simplifications, including the assumption of ideal stage efficiency, the omission of liquid holdup and transfer losses, and the lumping of ambient heat losses into a single effective heating parameter.

4.10.2 Numerical Benchmarking

Despite these simplifications, the custom model's numerical framework demonstrated close numerical agreement with Aspen Plus, with both platforms yielding closely aligned results under identical NRTL thermodynamic parameters and operating inputs. The sensitivity analysis further clarified the system's operational boundaries, highlighting the fundamental trade-off between production rate and product enrichment: increasing the reflux ratio improves ethanol concentration but reduces volume, while increasing heating power has the exact opposite effect. Additionally, the analysis indicated diminishing returns in product enrichment beyond four theoretical stages. Finally, while adjusting the model to reflect the local Kathmandu atmospheric pressure only shifted predictions by less than 0.5%, incorporating this environmental variable remained a necessary step to ensure strict thermodynamic consistency regarding boiling behavior and ethanol–water VLE [10, 26].

4.10.3 Energy and Economic Considerations

For Run 3, the 1025.3 W heating power applied over 53 minutes yielded a total energy input of roughly 3.3 MJ, with about 1.1 MJ rejected to the cooling water during collection. The remainder accounted for vaporization, sensible heating, and ambient heat losses. Constructed from locally available materials, the prototype offers a potentially cost-effective option for small-scale batch distillation. This experimental setup uses heat-exchanger reflux control, which avoids the use of expensive flowmeters, offering a cost-effective and accessible alternative for small-scale operations while maintaining practical predictive capability. For instance, in runs without observed leakage, deviations ranged from 5.62% to 10.42% for final concentration and 0.96% to 4.16% for collected volume, suggesting a general alignment between the model and experimental outcomes. Additionally, incorporating the local atmospheric pressure into the model helps account for anticipated pressure effects on ethanol–water VLE [10] and the related changes in distillation energy requirements [26].

5. CONCLUSION

This study developed and experimentally evaluated a low-cost, heat-exchanger-based reflux batch distillation system alongside a custom numerical modeling framework. The model, integrating NRTL-based equilibrium, McCabe–Thiele stage analysis, and local atmospheric

pressure correction, followed the transient distillation behavior without requiring direct measurements of internal vapor and liquid flow rates. For the primary experimental baseline (Run 3), the model predicted final distillate concentration (83.70% vs. 79% v/v) and volume (1094.49 mL vs. 1049 mL) with relative deviations of 5.62% and 4.16%, respectively. The model framework was further benchmarked against Aspen Plus, demonstrating numerical consistency with maximum deviations of 2.91% for composition and 5.59% for total distillate amount across all runs.

Furthermore, sensitivity analyses illustrated the fundamental trade-off between product purity (driven by higher reflux) and total yield (driven by higher heat input), demonstrating diminishing returns in separation efficiency beyond four theoretical stages. Finally, while correcting for the local Kathmandu atmospheric pressure altered final volume and composition predictions by less than 0.5%, incorporating this environmental variable ensured a consistent thermodynamic basis for the system's boiling behavior and energy input.

6. LIMITATIONS AND RECOMMENDATIONS

The current study's evaluation is limited by its assumption of ideal theoretical stages, the estimation of effective heating power and reflux ratio rather than their direct measurement, the determination of initial component inventories from measured feed volume and concentration using a mixture-density correlation, unquantified ambient heat losses, and the small number of experimental runs subject to minor cooling-water fluctuations.

To address these limitations and improve system performance, future work should focus on the following areas:

- Experimental Upgrades: Future setups should directly measure vapor and liquid flow rates, instantaneous distillate volume, and boiling-point evolution to better validate the dynamic material and energy balances. The physical apparatus can be improved by adding proper insulation to minimize thermal losses, implementing automated reflux control for operational consistency, and testing the system under alternative stable heat sources.

- Computational Refinements: The numerical model should be upgraded to incorporate actual tray and column efficiencies rather than assuming ideal stage behavior. Performance should also be benchmarked against conventional reflux configurations to evaluate the specific advantages of the heat-exchanger-based arrangement.

- Machine Learning Integration: The developed physics-based framework can serve as a data generator to train lightweight surrogate models across varied operating conditions. Such data-driven approximations could enable rapid, real-time composition estimates for low-cost monitoring without numerical overhead.

7. ACKNOWLEDGEMENTS

We would like to express our sincerest gratitude to the Department of Mechanical and Aerospace Engineering at the Pulchowk Campus, Institute of Engineering, Tribhuvan University, for providing us with access to the necessary laboratory facilities and equipment during the

construction and testing phases of our project. We owe special thanks to Vishwo Prasanna Amatya, and to the dedicated lab staff for providing invaluable guidance and unwavering support throughout all aspects of the fabrication and testing process. We are also deeply grateful for the generous financial support from the Free Students' Union 2079, without which this project would not have been possible. Finally, a huge thanks to our project teammates, Sanjiv Pahari and Madan Ghimire, along with all our friends and classmates who assisted with fabrication and testing and provided us with consistent encouragement from the initial design conception through our final experimental run.

REFERENCES

[1] Perry RH, Green DW, eds. Perry's Chemical Engineers' Handbook. 7th ed. New York: McGraw-Hill; 1997.

[2] Sholl D, Lively R. Seven chemical separations to change the world. Nature 2016;532:435–7. doi:10.1038/532435a.

[3] Sorensen E. Design and operation of batch distillation. In: Gorak A, Sorensen E, eds. Distillation: Fundamentals and Principles. London: Academic Press; 2014, p. 187–224. doi:10.1016/B978-0-12-386547-2.00005-3.

[4] Mujtaba IM, Macchietto S. Optimal operation of multicomponent batch distillation-multiperiod formulation and solution. Comput Chem Eng 1993;17:1191–207. doi:10.1016/0098-1354(93)80099-9.

[5] Krell E. Handbook of Laboratory Distillation. 2nd ed. Amsterdam: Elsevier; 1982.

[6] Ye L, Zhang N, Li G, Gu D, Lu J, Lou Y. Intelligent optimization design of distillation columns using surrogate models based on GA-BP. Processes 2023;11:2386. doi:10.3390/pr11082386.

[7] Bonsfills A, Puigjaner L. Batch distillation: Simulation and experimental validation. Chem Eng Process Process Intensif 2004;43:1239–52. doi:10.1016/j.cep.2003.11.009.

[8] Alhaboubi NA. Constant distillate composition of a batch distillation column with variable reflux mode based on still pot concentration. Sci Tech Energ Transition 2022;77:2. doi:10.2516/stet/2021002.

[9] Hála E, Pick J, Fried V. Vapour-Liquid Equilibrium. Amsterdam: Elsevier; 2013.

[10] Guevara Luna MA, Guevara Luna FA, Belalcázar Cerón LC. Experimental data and new binary interaction parameters for ethanol–water VLE at low pressures using NRTL and UNIQUAC. Tecciencia 2018;13:17–27. doi:10.18180/tecciencia.2018.24.3.

[11] McCabe WL, Smith JC, Harriott P. Unit Operations of Chemical Engineering. 5th ed. New York: McGraw-Hill; 1993.

[12] Diwekar U. Batch Processing: Modeling and Design. Boca Raton: CRC Press; 2014.

[13] Luyben WL. Aspen Dynamics simulation of a middle-vessel batch distillation process. J Process Control 2015;33:49–59. doi:10.1016/j.jprocont.2015.06.002.

[14] Ortega J, Fernández L, Sosa A, Lorenzo B, Ríos R, Wisniak J. New advances in the modeling and verification of experimental information for ester–alkane solutions: Application to a batch-distillation case. Ind Eng Chem Res 2020;59:8346–60. doi:10.1021/acs.iecr.0c00850.

[15] Schwarz CE. Effect of variation between different experimental VLE data sets on thermodynamic model and separation predictions: NRTL correlation of the ethanol + water system. Ind Eng Chem Res 2024;63:10721–34. doi:10.1021/acs.iecr.4c00484.

[16] Renon H, Prausnitz JM. Local compositions in thermodynamic excess functions for liquid mixtures. AIChE J 1968;14:135–44. doi:10.1002/aic.690140124.

[17] Abrams DS, Prausnitz JM. Statistical thermodynamics of liquid mixtures: A new expression for the excess Gibbs energy of partly or completely miscible systems. AIChE J 1975;21:116–28. doi:10.1002/aic.690210115.

[18] Smith JM, Van Ness HC, Abbott MM. Introduction to Chemical Engineering Thermodynamics. 7th ed. New York: McGraw-Hill; 2005.

[19] Li J, Chen C, Wang J. Vapor–liquid equilibrium data and their correlation for binary systems consisting of ethanol, 2-propanol, 1,2-ethanediol, and methyl benzoate. Fluid Phase Equilib 2000;169:75–84. doi:10.1016/S0378-3812(99)00340-4.

[20] Do XT, Nguyen THT, Pham NT. Experimental study and process modeling of a vacuum batch rectification column for cinnamon essential oil fractionation. Food Bioprod Process 2024;148:400–9. doi:10.1016/j.fbp.2024.10.009.

[21] Niazi S, Díaz-López JA, Nieto-Márquez A. Improvement of energy efficiency and production performance in a heteroazeotropic batch distillation unit: A study on decanter control and feeding strategy. Sep Purif Technol 2025;357:130132. doi:10.1016/j.seppur.2024.130132.

[22] Bubel M, Seidel T, Bortz M. Reusable surrogate models for distillation columns. Comput Chem Eng 2026;206:109523. doi:10.1016/j.compchemeng.2025.109523.

[23] Chen H, Flores GEC, Li C. Physics-informed neural networks with hard linear equality constraints. Comput Chem Eng 2024;189:108764. doi:10.1016/j.compchemeng.2024.108764.

[24] Staud R, Schröder K, Mesarek K, Schröder J, Herbst R, Jasch K, et al. Impact of thermal column wall conditions and subcooled reflux on separation performance in miniplant distillation columns. Sep Purif Technol 2025;358:130479. doi:10.1016/j.seppur.2024.130479.

[25] Ränger L-M, Waibel Y, Grützner T. Experimental investigation of heat losses in a pilot-scale multiple dividing wall distillation column with three parallel sections. ChemEngineering 2023;7:68. doi:10.3390/chemengineering7040068.

[26] Duan W, Yang M, Feng X. Comprehensive analysis and targeting of distillation integrated into overall process considering operating pressure change. Processes 2022;10:1861. doi:10.3390/pr10091861.

[27] Battisti R, Claumann CA, Marangoni C, Machado RAF. Optimization of pressure-swing distillation for anhydrous ethanol purification by the simulated annealing algorithm. Braz J Chem Eng 2019;36:453–69. doi:10.1590/0104-6632.20190361s20180133.

[28] Rougier M, Bellettre J, Luo L. An experimental study of a wine batch distillation in a copper pot still heated by gas. Energies 2021;14:3352. doi:10.3390/en14113352.

[29] Barolo M, Botteon F. Calculation of parametric sensitivity in binary batch distillation. Chem Eng Sci 1998;53:1819–34. doi:10.1016/S0009-2509(98)00017-7.

[30] Yanbolagh DJ, Saraei A, Mazaheri H, Mehrabadi SJ. Exergoeconomic, environmental, economic, and energy-matrices (4E) analysis of three solar distillation systems equipped with a condenser and different heaters. J Therm Eng 2021;7:1640–53. doi:10.18186/thermal.1025916.

[31] Kumar S, Chandra P. Experimental investigation of axial finned tube evaporator thermal distillation system for diesel engine waste heat recovery process. J Therm Eng 2024;10:479–89. doi:10.18186/thermal.1456708.

[32] Douady A, Puentes C, Awad P, Esteban-Decloux M. Batch distillation of spirits: Experimental study and simulation of the behaviour of volatile aroma compounds. J Inst Brew 2019;125:268–83. doi:10.1002/jib.560.

[33] May-Vázquez MM, Rodríguez-Ángeles MA, Gómez-Castro FI, Espinoza-Zamora J, Murrieta-Luna E. Development of a mass transfer model for the rate-based simulation of a batch distillation column. Comput Chem Eng 2020;140:106981. doi:10.1016/j.compchemeng.2020.106981.

[34] Pacheco-Basulto JÁ, Hernández-McConville D, Barroso-Muñoz FO, Hernández S, Segovia-Hernández JG, Castro-Montoya AJ, et al. Purification of bioethanol using extractive batch distillation: Simulation and experimental studies. Chem Eng Process Process Intensif 2012;61:30–5. doi:10.1016/j.cep.2012.06.015.

[35] Rérat C, Papadokonstantakis S, Hungerbühler K. Estimation and analysis of energy utilities consumption in batch chemical industry through thermal losses modeling. Ind Eng Chem Res 2012;51:10416–32. doi:10.1021/ie2000693.

[36] Gmehling J. Vapor–Liquid Equilibrium Data Collection. DECHEMA Chemistry Data Series, Vol. 1. Frankfurt am Main: DECHEMA; 1977–2008.

[37] Osborne NS, Stimson HF, Ginnings DC. Measurements of heat capacity and heat of vaporization of water in the range 0 to 100 °C. J Res Natl Bur Stand 1939;23:197–260.

[38] Miyazawa T, Kondo S, Suzuki T, Sato H. Specific heat capacity at constant pressure of ethanol by flow calorimetry. J Chem Eng Data 2012;57:1700–7. doi:10.1021/je2013473.

[39] Gaulhofer A, Kolbe B, Gmehling J. Thermodynamic properties of ethanol and water—III. Description of the different excess functions (gE, hE, cEP) using an empirical model. Fluid Phase Equilib 1988;39:193–209. doi:10.1016/0378-3812(88)85004-0.

[40] Washburn EW. International Critical Tables of Numerical Data, Physics, Chemistry and Technology. Vol. 3. New York: McGraw-Hill; 1928.

[41] Lange NA. Lange's Handbook of Chemistry. 15th ed. New York: McGraw-Hill; 1999.

[42] JCGM. Evaluation of measurement data – Supplement 1 to the "Guide to the expression of uncertainty in measurement" – Propagation of distributions using a Monte Carlo method. JCGM 101:2008. Sèvres: Joint Committee for Guides in Metrology, BIPM; 2008.

[43] Ashraf WM, Dua V. Partial derivative-based dynamic sensitivity analysis expression for non-linear autoregressive with exogenous (NARX) model case studies on distillation columns and model's interpretation investigation. Chem Eng J Adv 2024;18:100605. doi:10.1016/j.ceja.2024.100605.

[44] Aspen Technology, Inc. Aspen Plus V14. Bedford, MA: Aspen Technology, Inc.; 2022.

[45] Zhang C, Liu Y, Gao Z, Huang L, Xiong J. Process design, simulation and experimental studies of heteroazeotropic batch distillation for phenol dehydration. Chem Eng Res Des 2023;195:682–90. doi:10.1016/j.cherd.2023.05.062.

CODE AVAILABILITY

The Python scripts and related materials used for the numerical modeling and analysis are publicly available at the GitHub repository:
https://github.com/basnetsaroj/Batch_distillation_paper